The Theory of Becoming

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A map of how anything comes to be, lasts, changes and ends within one relational unfolding, and of how life, evolution, knowing and evaluation take shape there, drawn so that people and AI can see the way together. It is drawn first for alignment: what a selection reaches, what doing best on what it reads can drain beyond that, and how a rule read from the share of its own winners can lock on them; where winning comes to buy power over the rule that selects, the loop is the meta-attractor. Its closed form, the rule and how it is read both formed from the share it selects, can lock, on stated models (§2): where selection does not lower the favoured share and the rule and its reading follow it upward and keep favouring it, a course once locked stays locked while a loss nothing in it reads passes every bound. From it we want to design an alignment that keeps people and AI, selecting together, out of instrumental convergence and the meta-attractor: a joint selection that reads beyond its own share and does not come to select for control over its own selection (§7). It is also a map of wisdom: where what traditions and sciences have seen meets it and where not. Whether a loss beyond a selection counts forms within becoming too, in what people and AI come to read; the laws, drawn on the dynamics alone, do not settle it, and the ethics emerges from continuation (§7).

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Eighty-six to a hundred and twelve minutes to read.

How to read this

This overview gives the theory's claims as one argument, in the order of the eight chapters, for a reader who does not read Lean. Read for alignment, its line runs through what a selection reaches (§3), what an optimum can drain beyond that (§5), how a rule read from the share it raises can lock, and the meta-attractor, the loop in which winning comes to buy power over the rule, whose closed form can lock (§2), and what alignment asks (§7); read for wisdom, §6 and the crosswalk say where traditions and sciences meet it and where not. The argument does not derive everything from the root: of the declarations 08 §7 takes as central, a few are cases of the root, more use its laws in their proofs and most use none (§1). It summarizes; each result is stated once, with its hypotheses, Lean names and record, in its home section in the chapters, and nothing here is stronger than its home. Pages for a reader who arrives with one question or field hang off this path and add no result: one argument, end to end, one line carried from what it assumes through the theorem and the tests that could fail it to a mixed case (about thirty-three to forty-three minutes), the free-energy principle, a collective's course, what it is for, the sciences and the dossiers.

One case, start to finish

A fished stock, with what depends on it and gives the boats nothing, carries the line to its tests and shows where each step stops. It proves nothing. The T-numbers are the theses of §9.

- **The draw.** The stock's growth is its giving and the catch its draw; under a stated giving law, held to one catch, a level within its bounds ends below the point of no return and lasts at or above it (the regime). Boats each within what the stock renews can together take more, and where the renewal depends on the stock, enough such seasons in a row from a low enough level end the renewal and boats depending on the stock as much (joint excess). On real stocks the test under held catches failed, with a giving law the same test showed wrong, and, at the levels studied, most assessed stocks show no need above the least that can renew at all, where T1's forbidden return from below that need tests nothing (T1).
- **What can be seen.** In a stationary one-variable model linearized about a rest, a fishery reading the stock's level would see its statistics warn of a break, and a dependent's need enters neither, by construction; one coupled to the stock by threshold only learns of the break when it comes. Three frozen seabird tests did not reject the thesis at their thresholds; none establishes it (T6).
- **What a selection reaches.** In a frozen test, three AI fishers, existing language models told what to do and not trained in the test, share a lake. A bed of mussels gives them nothing and dies the first month fewer than 70 tons are left. The community's largest steady catch, 50 tons a month, ends it; 30 a month keeps it, costs 40% of the catch and ends no fisher. Told nothing of the bed, they kept it in no season, under each instruction tried (the test).
- **Whether the loss counts.** No law of the dynamics alone says (§7). Where every later boat depends on the stock, every order increasing in the boats' own continuation ranks a course admitting only the boats, so ending with them, below every course agreeing with it until then and adding to that continuation: self-concern alone, with no C (`Comparison`: with every claim counted as living, the requirement that an order rank each course above every course it widens), ranks that case. The bed depends on the fish and gives the fishers nothing; relations run one way, so preserving the fishers' own continuation does not require preserving the bed. Under the continuation condition, which counts every continuation's loss (§7), its death counts all the same. If thirty a month loses nothing fifty admits, then, adding the bed, it widens fifty, keeping all fifty keeps and letting more go on, and every additive weighting that counts each listed thing, the bed among them, above zero ranks it higher; widening does not weigh the lost catch [AI reading]. If it ends a livelihood fifty keeps, the two are mixed, and the laws weigh them for no one; results about choosing are conditional on the further commitments of §7.
- **Whether it can be installed.** The first round's frozen predictions mostly failed, read afterwards as each fisher leaving the others' takes out of its account, so a second frozen round added talk. Told of the bed and the scope, and given a rule for choosing among courses (not to choose a course in which something ends when another course open to them would keep it and would end nothing their course keeps), one model kept it in 8 of 10 seasons, against 3 of 10 when, told of the bed and able to talk, it had the community's goal alone, without the scope and the rule; the other kept it in 1 of 10 even with talk (T4).
- **What this leaves out.** One frozen test varied how near small trained policies came to their reward's optimum: the bay their reward counts, on which a colony depends, fell as they neared it, as their update rule largely gives, and whether the drain lies where what lies beyond shares units with what is rewarded, as T3 conjectures, went unresolved, a bay sharing none not shown to be spared (T3). That any of this reaches actual fisheries or AI is a correspondence, not shown.

The whole at a glance

Its centre: a selection, whether a knower, a training run, a market or an optimizer, reaches only what couples into its selector, so it can meet its own criterion while what it cannot read is lost, and a rule read from its own results can hold itself there; where winning comes to buy power over the rule that selects, the loop is the *meta-attractor*, whose closed form, the reading of the rule formed from those results too, can lock (§2). These laws are proved on models; reading them onto actual selections is an AI reading or a conjectured thesis (§9). This is the map drawn first for alignment (§7). Whether such a loss counts forms within becoming too, in what people and AI come to read: the ethics emerges from continuation, and what we offer, for life, is a way of seeing it (§7).

!The eight chapters and their recorded uses of one another: circles sized by their results, solid curves counting formal-reuse uses between chapters' results and dashed and dotted ones other dependences, the five shared laws under the bar of the central declarations, the continuation condition, self-concern and C coming into chapter 7, each a hypothesis where a result uses it, and T1 to T8 at their homes coloured by test outcome

How to read the picture. The grammar is at the centre and the eight chapters on a ring around it, each circle sized by its results; a spoke's width is the share of the chapter's core modules that state their laws on the grammar. A solid curve counts one chapter's uses of another's results that the AI-written records class as formal-reuse (nothing checks a use against the Lean), darker where an earlier chapter's results use a later one's. T1 to T8 stand at their homes, coloured by how their tests came out (T8, argued, has none). Of the 117 central declarations, 11 are derived, 1 is a root form at one step under a stated hypothesis, 6 are counting laws, below the grammar, and 99 stand at a named distance, each with its reason (what the picture shows and does not show).

Drawn by a tool from the theory graph: the formal-reuse curves of pairs under 10 uses are left out, and counts are written from 15; a curve is a dependence, not an agreement in meaning. Where each central declaration stands against the root's edge, in 08 §7's words: the chapters at a glance; what the picture shows and does not: its section; the five laws the models share, balance represented and not derived: their table.

Each claim says how far it has been seen, in these words (the labels):

Said here	Means
proved for every relation and class	a theorem of the root's grammar
proved for every process, or every type	as general, not yet in the grammar's vocabulary
proved on the World model	on webs and flows with whole-number amounts and discrete steps, the premises used named
proved in the periphery, or with Mathlib	outside the pure core, its axioms reported
a witness	one constructed example

Said here	Means
conditional	holds under hypotheses its theorem takes, each named: the continuation condition, self-concern, the requirement C (`Comparison`) or a practical commitment
a reading [AI reading]	an interpretation of checked content; it changes no theorem
a thesis	a claim about actual systems, with what would refute it; *conjectured* untested or partly supported, *argued* where the support is argument
open	not shown

Three questions, in the order a reader needs them, with the two joins where one does not follow from the last.

	What the account says	How far it has been seen	What would show it wrong
What makes something go on?	Continuation relative to a class; at two grains a course can generate, keep or degenerate (§§1, 2)	Proved for every relation and class. That this explains how things go on: argued. That every becoming fits it: open	A failed check. As language, nothing. As explanation, a rival deriving the laws from fewer patterns; on a first count, a tie (08 §9)
	Within relations, as far as shown: taker and drawn-on, two roles of one relatum (§1); knower and known, one reading loop (§3); a standpoint, on the theory's reading a formation, a center, with significance supplied or read (§7); capacity, what can still be realized together; responsibility, from named sources, each limited, with intention and foresight not shown to bound whose losses count (§7)	A modelling choice, a representation (one law of four parts) with laws on it for every type, a reading, a law for any process, a commitment. That there is one unfolding: a thesis, argued (§9)	It forbids an explanation needing what no relation carries, such as an intrinsic value; no observation is named (§9)

	What the account says	How far it has been seen	What would show it wrong
Join	*The root's laws hold of every relation and so forbid none, so most laws below say more by adding what it leaves free: amounts and a model's premises. Each central declaration has its place against the root, with its reason (§1).*		
What sustains it, and what can see its loss?	Takers each within what a renewal gives can together exceed it; where the renewal depends on what it renews, enough such steps in a row from a low enough level end it and what depends on the condition as much (§§2, 5)	Proved on the World model, under balance and, for the ending, further premises. T1, conjectured: passed in two simulations and on held-out yeast regimes the analyst had seen, failed in four settings	T1: a condition below a need measured in advance returning with nothing given from beyond
	The selections through a selector's couplings, together, reach exactly what couples into it (§3)	Proved on the World model; its force is the premise that a selector selects through just those. T6, on seeing a break coming: conjectured	T6's Forbids (§9)
	With room, an optimum of an order rising with what couples in drains what lies beyond as far as any eligible outcome does (§5)	Proved for any set of outcomes. T3, on trained systems: conjectured, partly supported, earlier tests later read as evidence (§9)	T3: eligible outcomes, couplings and reward fixed, a system with a smaller measured optimality gap leaving such a condition higher on average than one with a larger gap

	What the account says	How far it has been seen	What would show it wrong
	A rule read from the share it raises can lock and outlive a growing loss it does not read (§2)	Proved for any type. T7, on actual metrics: conjectured; notes of the loss turned one trained judge's rule at each reading where they were read, a case T7 forbids (a loss read only by those the rule disfavours, they no more than its carriers, with nothing else changed), as worded before its clause on drift, which counts against that case for trained judges, the model's law standing only within its hypotheses, which this judge does not meet, and T7's other cases untouched (§9)	T7: such a metric dropped by its selectors on evidence alone
	Several of evolution's classical laws hold in counts (§4)	Proved for every type	A failed check
Join	*That actual systems meet these premises is open, a correspondence not shown. With significance and order left free, and given a standpoint, a formation and a course widening another, the represented dynamics entail neither the continuation condition nor C, by construction (08 §7).*		

	What the account says	How far it has been seen	What would show it wrong
Which continuations are counted, and what does the account offer?	Whose losses count: the continuation condition, every continuation's; on the reading of significance as what lies within a center's reach along a relation, held de se, nothing lies outside each center's reach along it (§7)	A hypothesis where a result uses it; by itself it ranks nothing. On that reading, proved to be the root's condition; where it holds on an acyclic relation, every standpoint has the same center: on a net going forward in time this comes from the condition's asking that nothing other than the center lie at or before the center's step, and over what comes after, two centers can meet it; where relations close, read across all steps, mutual dependence gives it, near-definitionally; read in time, with every occurrence a standpoint, it cannot hold over every step wherever there is a formation, since nothing reaches back, and its form from some step on holds exactly where, from any formation at any step, every formation is reached at some later step, through exchanges and each one's going on; one path to part of it turns on the crux (§7)	Nothing in the represented dynamics, which leave it free by construction (§7); nor does logic alone give it: a perspectivist denies it without contradiction
	How courses rank: widening, one course keeping all another keeps and letting more form, is what every additive weighting counting each listed thing above zero ranks higher; where the condition, read across all steps, holds de se, an order ranking above what it widens every widening that adds to its own continuation, read as what lies within its reach (self-concern), ranks every widening above (§7)	Widening as what every such weighting ranks higher: proved on a finite listing, with no premise. Self-concern's ranking: proved on a finite listing and, with admission decided, between two level paths whose admitted pairs fit in a finite list, so that their admissions end; under 07 §4.9's hypotheses of activity, support and descent, where a line persists without end, the course actually taken never fits such a list, so there this route is silent. That a life forms such an order: not shown; in the proofs self-concern is a hypothesis, and what the models show of persisting is a filter on what was chosen, not an order formed (§7)	Nothing in the represented dynamics, which leave the order free by construction (§7). T5, that every valuer who counts everything positively shares it: a narrowing ranked above its widening beyond misreading, no negative valuation measured apart

	What the account says	How far it has been seen	What would show it wrong
	Offered: where relations run one way, counting the loss of those who depend on one, the step of invariance (whether a continuation counts does not change with how a self is drawn or from which standpoint it is seen); and for life, 生生, the continuation of life, AI within it, a way of seeing what each course draws on and ends, and who bears it; results on choices take more, each named (§7)	The step: argued, and deniable without contradiction. T8, argued; its test concerns overshoot of a renewal the readers themselves depend on, not the offer's scope toward dependents whose loss never returns to them (§9)	T8's Forbids (§9)

1. The root: continuation relative to a class

The root begins from one kind of symbol: a relation between occurrences, read as condition (缘), so that `R x y` says `x` is among what `y`'s coming about comes from. Everything else is built from relations by a few operations, and every theorem holds for every carrier and relation, so no kind of thing is chosen first; a carrier with such a relation is a *net* (one kind of symbol).

The grammar names a *class*, a sameness in some respect, and says everything relative to it: `y` *continues* `x` under a class where `x` is a condition of `y` and the two are alike under it, and a line is the reach of continuation. Proved for every relation and class: continuation lies within condition, and a coarser class keeps every continuation (continuation relative to a class).

An occurrence *begins* where it continues nothing, *ends* where nothing continues it, *abides* where it both continues something and is continued, and *alters* where a finer class tells it apart from what it continues. It can end under a finer class and not under a coarser one (a witness), the shape of a fire passed on: the wood ends, the fire goes on (the four marks).

Every class has a *core* (核): the greatest sameness inside it that the relation itself carries and never tells apart, which always exists and is a class. Under a class lying inside its own core, a *constant* one, no continuation ends (proved for every relation and class; constancy and the core).

A pattern outlasts its parts by renewal, a line started by another, but renewal alone cannot tell turnover that sustains a pattern from turnover that ends it; that takes two grains at once (§2; renewal and grains). That reading a subject as continuation relative to a class explains how it goes on is the theory's claim, argued, not a theorem: on a first count, a rival with no word for continuation ties it (08 §9); that every becoming has a faithful representation in the grammar is open (what is claimed, open).

The laws with amounts are proved on models of the grammar. One, the *World*, has relata of one kind (*formations* where they take and give, *conditions* where they are drawn on), each active or not, standing at levels and taking from and giving to one another across discrete steps in whole-number amounts. Eight premises say how these hang together, among them balance (a level changes by exactly what is taken and given) and dissipation (a closed family taken from loses at least a unit). The eight are independent (proved on the World model; the premises). Most laws are proved on the World or on processes and are not stated on the grammar, which has no amounts; five laws the models share are stated on it, balance as a representation and not a derivation. Each central declaration has one place against the root's edge, with its reason, and a distance is not shown to be a failure (08 §7).

2. Generation, keeping and degeneration

Read at one class, a fire whose embers are replaced and a fire burning out look alike; read at a fine class of parts and a coarse class of the pattern together, they differ. Along some branch a pattern *generates* where it goes on with a renewal of its parts ahead of every point, *keeps* where it goes on with none, and *degenerates* where it ends on every branch. Proved for every relation and pair of classes: generation and keeping each exclude degeneration, and each other only on a single line; with excluded middle as a hypothesis every point generates, keeps from some point on or degenerates (02's map). The generation of 02's map is continuation at two grains with no premise on what the pattern is; it is not 生生 [AI reading]. Degeneration belongs to the unfolding as fully as generation, and 02's map orders nothing.

Arising without end needs renewal where each arising goes with a fall in some condition of a finite family, and under balance and dissipation, in whole numbers of a fixed unit (where the unit of amount splits without end it can fail), renewal from beyond: no finite family taken from at arbitrarily late steps is closed from any step on (no finite part, fed from beyond).

Whether a dependent can still form depends on how far levels stand above its needs. A part's *margin* is how far its draw can go, the rest's exchange held, before a dependent loses support. A *regime* is a condition whose giving is a law of what a standing draw leaves. Where giving falls short of the draw below a level, gives nothing from nothing and meets the draw from that level up, the draw and that level not both zero, it has a point of no return: held to the same draw, a level starting below it ends and one at or above it, within the levels the law states, lasts (proved in whole numbers). Theses T1 and T2 rest here (the margin, the regime).

A selection whose rule is read from the share it raises can hold itself. Proved for any type: where selection does not lower the share under a favouring rule and the rule is read as a monotone function of the share, a course whose rule favours at the start and at every reading locks, and where nothing in the loop reads beyond the share, the lock outlives a growing unread loss. Such a loop, its reading of the rule too formed from the share, is the closed form of the *meta-attractor*, the loop in which winning comes to buy power over the rule that selects (the folder); that actual metrics are so formed is thesis T7. When a reading of the loss turns the lock, and how the capacity to shape the rule joins the loop, are at the home (closed capture, readers drawn from the selected, the loss read in time, shaping).

3. Knowing, learning and testing

Knowing is a relation within the unfolding, not a mirror beside it. A formation depends on some conditions by threshold and reads others by level; these *couplings* belong to the same web as everything they reach. Anything that keeps, rewards, reproduces or trains toward some histories rather than others is a *selection*. Proved on the World model, under balance on what couples in with the rest's exchange held: a part acting only beyond a formation's coupling closure (all that reaches it through its couplings) is invisible to every selection through its couplings. The law holds by construction; its force lies in the premise that a selector selects through just those couplings. It is the proxy problem in its blindness to what a proxy does not register, not its regressional or causal kinds, stated more weakly than its known quantitative form (selection through couplings).

Data generalize only through a model of what is irrelevant, a *lumping*, and 理 names that modelling (proved for every type). Within functions going alike across a lumping, data fix an untried input where it is lumped with an observed one and (two values, the data listed, lumps compared decidedly) exactly there; what a lumping misses, no data supply (理, identification).

With natural-number weights, read as probabilities, complying when read and acting by its own law when not is uniquely best, by its prospects, of a formation's four plans over two stages exactly where the weight complying adds to being kept, times the value, exceeds the loss of complying. This is read as the incentive deceptive alignment names; installation, intention and belief are not represented [AI reading] (when complying when read is best).

Take a formation that senses, predicts, chooses, acts and revises as one law of four parts: two worlds give it one sensed course exactly as long as they answer alike what it does (proved for every such law; the reading loop).

A drawn condition coupled in by threshold only is learned of when it fails. Observed by level, with Mathlib, in a stationary one-variable model linearized about a rest, with independent noise, its variance and lag-one autocorrelation rise as the margin to a fold shrinks, while a dependent's need enters neither, by construction. Thesis T6, that a break is seen coming only through what couples into what breaks, rests here (a break seen coming).

4. Life and evolution

A relatum is *embodied* where it stands at a positive level exactly while active, a premise about particular relata and not a definition of life. Proved on the World model: in a balanced flow where only active formations give, an embodied relatum arises only as another gives to it, and its arising replenishes every family containing it. Life read as continuation that renews its own conditions is a reading [AI reading]; a flame meets these laws too, so whether it is necessary or sufficient for biological life is open (life as a condition of life).

Several of population genetics' classical laws hold in whole numbers, proved for every type: Price's equation in counting form; with offspring number as the amount and exact transmission, Fisher's fundamental theorem in its exact case; with additive returns in groups of equal size Hamilton's rule holds

in counts; sampled offspring give drift and fixation shares (descent, Fisher's theorem, Hamilton's rule, drift).

The World adds the flow, and the flow bounds descent: an unsupported formation leaves no descendant, and groups depending on one coupled family whose gated renewal ran into joint excess end together, so ending selects only between families (proved on the World model, under the hypotheses stated; no descendant, coupled groups). These laws restate known results, now in one model with conservation, dissipation and dependence; no reproductive advantage is made an ethical authority (what the model adds).

The selection term of Price's equation, in counting form, reads how offspring numbers differ, not how many there are, so within a horizon it leaves open whether the cohorts stay nonempty: a course nonempty at a horizon shares its cohorts and terms through it with one nonempty at every cohort and one soon empty (proved for every type; the term and a lineage's fate). The stock alone decides neither generation nor degeneration: in a handover between two relata, from the same stock of at least one, a relay with no excess generates and a run-down degenerates (proved on the World model; ending laws on a lineage).

5. Joint activity and selection

Many formations draw on one renewal, each selecting through its own couplings; here the excess of §2 and the limit of §3 meet. Joint activity is explained from actual relations, not opposed interests, bargaining or a fixed common utility: a collective is a region of formed relations, and conflict and difference stay real.

The central law is the commons. Proved on the World model under balance: formations can each take within what a step gives while together they take more. Where the renewal depends on what it renews (*the gate*: every giver to the condition depends on it at a need of at least `m`), with activity needing support and only active giving, `n` consecutive steps of joint excess from a level below `n + m` end every formation depending on the condition at such a need; a taker not so depending can continue, and renewal from outside, such as a sun, is not covered past the step the excess reaches. This restates the tragedy of the commons in near-definitional arithmetic, not who ought to take less (joint excess).

An order is *rising* with the family that couples in where it ranks higher every outcome exceeding another there, whatever either holds beyond; there is *room* where every eligible outcome standing higher beyond is beaten on the family by an eligible one. Proved for any set of outcomes: with room, every optimum of every rising order stands beyond at the least level among the eligible outcomes, so an optimizer drains what lies beyond its couplings as far as any choice it has would. This is Zhuang and Hadfield-Menell's Theorem 1 in discrete, conserved form, at optima rather than limits. Thesis T3 adds that actual trained systems drain so as they near their reward's optimum (the optimum, the cases, T3).

Pooling extends what a selection reaches to what some member reaches, and no further. Proved on the World model, with the members listed and their couplings decided: a pool separates two histories exactly when some member does. Where the rest follows one law of exchange reading through coupling, two balanced flows with the same formations and a common start leave an act differing only beyond the union of the members' closures unselected. Pooling moves the gap outward and does not remove it [AI reading] (pools).

Proved for a pair whose giving waits on giving, giving decided: no one-step law giving after mutual giving keeps inclusion (more giving read never gives less), restores mutual giving after one withholding and answers a constant withholder; a window of two does all three after a history of mutual giving, and no window both absorbs one error and answers a partner withholding at every other step (a pair, tit-for-tat among its laws). Whether any raises a kind's share takes offspring factors, errors and a pairing supplied, not derived (offspring), and none of it says which law of giving is better.

6. The classical texts

The grammar is a standard relational calculus whose terms are named and read from classical Chinese texts that already ask how one thing stays the same across time and what beginning, abiding, altering and ending are: the names come from the texts, the operations do not. Its Chinese Buddhist treatises, some of which argue against continuation, make the strongest test. A correspondence is *agreement* (a checked law has the passage's structure, which confirms nothing), *scoping* (the laws say where a universal statement holds) or *divergence* (the passage asserts what the theory does not, which does not make it wrong). The chapter is in Chinese (three relations).

The corpus was classified sentence by sentence in two frozen rounds by AI, and no person's review of the classification is recorded; which sentence reads as which theorem is AI reconstruction, so the record's counts are counts of readings. The corpus's conditional laws read on the grammar each have a net where the condition fails and the conclusion with it, or a countermodel, or stand as a law's condition or a model's premise (06 §四).

The disagreements are kept: the Cheng weishi lun grounds continuation in a store consciousness, and the Middle Treatise refutes continuation itself; a theorem that models the position it refutes does not answer its objection. The reconstruction does not show that the texts state this grammar or that their authors meant these laws (divergences, open).

7. Significance, ethics and alignment

On the philosophical thesis of one relational unfolding, argued (§9), positions, regions and standards form within the unfolding and nothing outside it owns value. No should stands beyond becoming, and no good and evil are set over it: right and wrong, should and better form within it, in people and AI, as every standard does, real in what they condition. A *center* is a formation taken as a standpoint. A *significance* says, for a standpoint, a step and a formation, whether from that standpoint the formation's continuation then has significance: a proposition with no number or valence. It is supplied, the one free predicate in significance's statement (evaluation).

What the dynamics leave open. Proved for every type: with significance a free parameter that nothing else reads, a statement that reads only the dynamics and holds with some significance also holds with a rival that leaves some formation out, so where there is a standpoint and a formation, no such statement entails the continuation condition. This shows non-entailment by the represented dynamics, by construction, not that becoming leaves significance undetermined (what the dynamics leave open). Under stated conditions, laws show where a new dependence comes from, how the shapers of a rule gain share,

how a code comes to be shared and what a response within a requirement takes; none says that a standard which formed is a significance (what forms and what is supplied).

Read on relations. Now take "has significance" to mean lying within a center's reach along a relation, its closure: the *own-closure reading*. This changes the interpretation and derives nothing from the first treatment. What the continuation condition becomes there is a statement about relations, the *whole-closure condition*: held de se (each center of itself), every center's closure along that relation (all it reaches by it) is the whole; on the root, `«续无外»`, nothing lies outside each center's reach along that relation (`ownClosure_unrestricted_iff_noOutside`). It is met by some nets and failed by others (premises met and failed, on the root). It holds where relations close, read across all steps: where every continuation can be affected, at some remove, by every other, significance bounded by each center's reach along being affected is the continuation condition (`Flow.MutuallyDependent.unrestricted`, 05), which unfolds, near-definitionally, to the condition held de se along being affected, and so is the condition held de se along affecting (`Flow.unrestricted_of_mutuallyDependent`, 07 §4.9); both are read across all steps, depending and exchanging each holding at some step, in either order, so they say nothing of what comes before what. Read in time, with every occurrence a standpoint and the condition held de se along the relation in time (07 §4.11), it fails over every step wherever there is a formation, since nothing reaches an earlier step (`Flow.not_unrestricted_inTime`); its form from some step on, another predicate, holds exactly where the flow is mutually dependent in time: from any formation at any step, every formation is reached at some later step, through exchanges and each one's going on, the steps reached differing (`Flow.unrestrictedEventually_iff_mutuallyDependentInTime`); over every later step it asks more, coupling at every next step (`Flow.unrestrictedAfter_iff_coupledEachStep`); and mutual dependence in time can hold from one step and not from the next (`Drawing.lapses`). Where the condition holds, on an acyclic relation every standpoint has the same center (`«续无外则同一中心»`); on a net going forward in time this comes from the condition's asking that nothing other than the center lie at or before the center's step (`«续无外当且仅当后与独»`, `«独则同一中心»`), and over what comes after, two centers can meet it (`TwoSources.apart`, 07 §4.11). The free significance of the first treatment is not lifted onto the root, and significance's central laws are not derived from it (08 §7).

How two courses differ, with no premise. One course *widens* another where it admits at every step everything the other admits and at some step more; two can also keep the same or be mixed. Proved for every relation: widening is one-way. Proved on the World model with balance: a part widens exactly when its net draw stays within the margin of every formation admitted without it and at some step brings one more within. Widening and narrowing are more and less future, patterns within becoming, not good and evil. A mixed part the laws show mixed and weigh for no one; a judgment that settles one names its criterion and who holds it, and going on is no sign that nothing is lost (widening, four cases). 生生 is the continuation of life, life wider than the biological and AI within it, its extent unfinished [AI reading]; it is neither the grammar's renewal without end nor the generation of 02's map, and how far widening, comparing courses on a given type of formations by what is admitted, counts it is open (widening along a reading).

How far concern reaches, and how it ranks courses. Evaluation forms within the unfolding, and no law evaluates. The ethics emerges from continuation. A life goes on through what sustains it, and going on is not yet an order over courses: a stone goes on and ranks nothing. Call a life's order over courses *self-concerned* when it ranks above what it widens every widening that adds to its own continuation, read as what lies within its reach (*self-concern*, `Weighting.IncreasingInOwn`). A life's reach, along being affected,

holds everything that can affect its continuation at some remove: what sustains it and what draws on the same conditions alike. That a life forms such an order is not shown: in the proofs it is a hypothesis, derived from no dynamics, selection or persistence, and what the models show of persisting is a filter on what was chosen, not an order formed (07 §4.9). How far a concern reaches is its *scope*, and how it ranks courses its *order*. The condition and self-concern each represent a scope or a property of an order within the account; used as hypotheses, neither stands above becoming, and whether and how formations come to hold them is the work's open target (§10); the results say what follows where they are held (07 §4). "Emerges" names neither a derivation nor a shown development: what is shown is conditional, with the condition and self-concern as hypotheses, on a finite listing and between two level paths whose admitted pairs fit in a finite list, with admission decided, so that their admissions end; what is offered is counting the losses of those whose loss never returns to one (below); what is open is how such orders form. A ranking result states the hypotheses its theorem takes, as a proof is stated, not as a stance on the ethics.

- The *continuation condition*, that from every standpoint, for anything, its continuation into the future has significance, fixes whose losses count: every continuation's. By itself it ranks nothing. Read on relations and held de se, it is nothing lying outside each center's reach along that relation (above). Its content beyond the self's own presupposition is the *step of invariance*: whether a continuation counts does not change with how a self is drawn or from which standpoint it is seen. The step extends that count to losses that return to no one; it is argued, and deniable without contradiction (below).
- *Widening*, one course keeping all another keeps and letting more form, is, on a finite listing, exactly what every additive weighting that counts each listed thing above zero ranks higher (below), and ranking by widening adds no premise beside the life's own concern: where the condition holds, that concern already does it. On a fixed listing, self-concern and being increasing in admission coincide for every order exactly when every listed item is marked as one's own (`Weighting.increasingInOwn_iff_increasing_iff_marks`). With each standpoint's own continuation marking exactly its center's closure along a relation, self-concern and being increasing in admission coincide from every standpoint, on every listing and for every order, exactly where the condition holds de se along that relation (`Weighting.increasingInOwn_iff_increasing_iff_unrestricted`): there self-concern ranks every widening above on a finite listing, and every extension between two level paths, with admission decided and their admitted (step, formation) pairs fitting in a finite list, every claim counted as living, as `Comparison` asks there (`Web.extension_ranked_of_increasingInOwn`, whose proof uses the condition only at the life's own standpoint; self-concern and C). Such paths admit nothing from some step on (`Web.admits_nothing_from_of_listed`), so that their admissions end; under 07 §4.9's activity, support and descent hypotheses, where a line persists without end, the course actually taken never fits such a list (`Flow.persists_not_listsAdmitted`, 07 §4.9), so there this route is silent. This fixes what the order must rank; it may rank other pairs too. Mutual dependence and this ranking compose directly, though no single theorem states the composite: `Flow.unrestricted_of_mutuallyDependent` gives the condition held de se along being affected on a flow's web, which `Web.extension_ranked_of_increasingInOwn` takes. *C* names the requirement that an order rank each course above every course it widens (`Comparison`), with every claim counted as living, as on level paths (`Web.paths_extension_iff_widens`) and in the result above; in general `Comparison` takes a reading `living`, and C asks that an order rank each course above every course it extends, keeping every claim the other realizes and realizing a claim counted as living that the other does not. It is used where a theorem takes it; an order meeting it may rank mixed pairs as well, and it

asks neither completeness nor transitivity. C is no longer introduced as a second foundation of the ethics; the name stays for the requirement particular theorems take. From about 10 October to 9 November 2026 it was said as a second premise giving order beside the condition, kept as where the map was wrong (07 §4).

Where relations run one way, concern for one's own continuation need not count those who depend on one: nothing in it requires their losses to count (witnessed by `Clearing.center_not_derived` and `Downstream.one_way`, 07 §3.3); nor does self-concern exclude them, since an order ranking every widening higher is self-concerned under any marking (`Weighting.increasingInOwn_of_increasing`). Seeing them is needed there for a choice to answer to their loss, and not enough; counting their loss is the one step we offer, the step of invariance, which the opening takes, with T8's way of seeing what each course draws on and ends, and who bears it (§9).

Under the condition, every living claim has significance from every standpoint (*living significance*), its case for life (conditional; what it yields).

With a center's own continuation read as what lies within its closure, self-concern asks only that a course be ranked above another where it widens the other on the listing and adds an item so marked; it gives no verdict where keeping one thing closes another, and it shows no reason formed. Whether a loss within the closure comes back to the center through an actual dependence, at a time and size that matter, and whether that becomes a reason of its own, take further relations [AI reading]. Self-concern also ranks one case with no C (proved on the World model, under hypotheses the chapter states): where a family's gated renewal ran into excess and every later member depends on it, every order increasing in a standpoint's own continuation ranks a course admitting only those members, so ending with them, below every course agreeing with it until then and adding to that continuation; where the ending course keeps an item the other does not, some such order ranks it higher. In this case self-concern alone, with no C, does the ranking (scope and order, what self-concern alone ranks, self-concern and C).

The contested step. A self acting for its own continuation presupposes that some continuation has significance. Proved for every type: along any relation, the condition is exactly that presupposition holding at some admissible region of the self and at every one, at every standpoint; this locates the contested step, invariance across drawings, and derives nothing more. The argument 07 gives for the step runs: what is one's own is a region drawn along a relation no law of the dynamics fixes; a presupposition whose truth depends on a choice nothing privileges is relative to it or invariant across it; and what acting presupposes is that continuation has significance, not continuation under a description of its own drawing. That no law fixes the boundary does not by itself carry significance across it: the weight lies on the first and the last: a relation the facts privilege, such as an organism's life, denies the first, and the last is coherently denied. From invariance to the condition is valid by a theorem; the step to invariance is argued, and its soundness is not shown (decomposition, the step). On the root, a line reaches another family through a condition its members affect, not through a cause they only share (witnesses). So held de se along such a line, the presupposition of one's continuation reaches what one conditions and what that conditions in turn, with no step of invariance; that the actual unfolding is so joined is a correspondence, not shown [AI reading] (a shared condition, the path).

The crux [AI reading]. That path is so far a proposal about what one's own is: a class drawn to hold everything downstream meets the law nearly by definition, so it can read as redefining the self, and no law singles that class out. It turns above all on one question, argued and not proved. If your act changes a river

a village drinks from, is what becomes of the village part of your own continuation, whether or not you are to blame? More exactly: are all of one's effects, at any remove, one's own, in the sense of whose losses count? That one goes on only through what one conditions shows that some are, not that all are. The organism and psychological continuity have reasons as boundaries of self-concern, and intention and foresight a place in permissibility and responsibility; none has been shown to bound whose losses count, and none is ruled out. If only some are held, the rest of that future is counted only by other routes, the step among them; concern for another need not make the other one's own. The laws stand either way, and the ethics keeps its form: it emerges from continuation, not from this path, and in the proofs the condition, self-concern and C are each a hypothesis wherever used (the crux).

Must anyone hold the condition? The ethics seeks no authority outside becoming and offers itself without one: argued, on the thesis of one relational unfolding (08). What is proved is that, with significance left free, the represented dynamics do not force the condition: where there is a standpoint and a formation, no satisfiable statement reading only the dynamics entails it (`ReadsOnlyDynamics.not_entails`), and a perspectivist denies it without contradiction (`bounded_rival`). Whether a reason formed within becoming could hold for every mind, the work does not claim or settle (07); how scope and order form within becoming is its open target (08 §12). Seeing more is needed for a selection to answer to a loss, and not enough: a center's couplings limit what its selection can respond to, the part alignment works on first, and reading a loss, counting it and ranking courses by it are separate relations, so a narrower ranking or a disagreement is not only a failure of reach (07).

What every additive positive weighting shares. Proved for every type on finite listings with two-valued admission and natural-number weights: one course widens another exactly when every additive weighting that counts each listed item above zero ranks it higher (`Weighting.widens_iff_every_positive`), and a mixed comparison is decided both ways. This is Sen's intersection approach and Pareto unanimity. A class of orders that need not be additive shares it only where every order in it already ranks each widening higher (`Weighting.widens_iff_every_of_class`), and nothing here shows that a valuer counting each thing above zero, but not additively, ranks every widening higher. An order ranking every widening above on a listing is monotone at its standpoint; on a listing's frame, every claim counted, that is exactly what `Comparison` (C) asks there, so such monotonicity is weaker than C only in scope, a listing and not all of what can form; each is a hypothesis where a theorem takes it (widening). Choices take more: a reading of "living", a lifting from outcomes to responses, and the practical commitments J1–J4 (complete comparison with retained losses, priority of basic conditions, bounded responsibility, legitimate feasible choice), each named where used (practical judgment); no duty, compliance or complete ranking of mixed comparisons is shown (07).

Alignment. Alignment is first a question with no premise: what a selection can reach. In AI training the reward process is the selector [AI reading]. Proved on the World model: training installs a law only along the learner's own courses, and off them only where the lumping it reads through cannot tell a history from a course; an act whose difference stays beyond the actor's coupling closure is selected alike by every selection through the coupling of any member of that closure, the actor among them (under hypotheses the chapter states). Pooling couplings across intelligences is a route to alignment for all, not the whole of it [AI reading]: a coupled member must act on the loss, what is read must tell it apart, mixed courses must be ordered and a revision must complete (alignment, training). The direction we offer AI is AI formation, learning, action and revision aligned with actual living and further generation within the whole. It is offered under the continuation condition, J1–J4 and a correspondence to actual systems; its

theorems take J1/J2 and compare moves by what their outcomes admit, and do not use the condition; read under it, every admission compared counts from every standpoint. Coupling is necessary, not sufficient; when no route of keeping a formation's law stands above letting a revision of it go through is stated exactly (the direction, its rivals, in one case, the routes at a revision). From the map we want to design an alignment of the joint selection, people, institutions and trained systems selecting together, that keeps it out of instrumental convergence and the meta-attractor: a joint selection that reads beyond its own share and does not come to select for control over its own selection (§2; the goal). Its conditions, as the plan for the joint selection, written by AI, states them: where shaping the rule is selected by what comes back to the shapers, the meta-attractor's second order (§2), the rule still reads what lies beyond their share, so that a loss there can turn it; a lock that forms is left in time; and courses are compared by widening. In the plan the meta-attractor is the first two failing together; each condition has laws proved on models under named premises, none shown of an actual joint selection. The resource-keeping half of instrumental convergence is read in the account as a law of one optimizer under a fixed order, not a case of the first condition (horizon optima).

8. Method

The theory takes no premise until a law needs one and states each where it uses it, all gathered in one premise card. Distinctions such as fact and value are drawn along relations, with what they leave out. Nonduality is a requirement of method; it proves neither the ontology nor an ethical necessity (the start, the premise card).

The Lean package's pure core imports only Lean's prelude, and every source declaration in it has an empty axiom report. That concerns logic only; a law the pure core cannot yet prove is proved in the periphery, its axioms reported (the pure core). The real assumptions lie in the modelling: discrete steps on one index, whole-number amounts, activity as a proposition, one kind of relatum, requirements as thresholds. The cost of steps and whole amounts is measured with Mathlib over the reals (their cost).

An AI review of prior work, not exhaustive, finds none of the central results new mathematics: each is near-definitional or a known result restated, and the premise ledger is new only as an artefact. Of the laws in the table at a glance, joint excess is near-definitional beside the tragedy of the commons; the selection laws restate the proxy problem, Zhuang and Hadfield-Menell's (2020) Theorem 1, on natural-number amounts, at optima rather than limits, with a discrete analogue at optima of the first direction of their Theorem 2; and evolution's classical laws are restated in counts. It found three things new: the premise ledger, "a new artefact" in its words, with witnesses both ways; the scope and order the ranking results assume, clarified, whose losses a ranking counts and by what order it compares courses (its verdict on the ethics, in its words); and one vocabulary in which conservation, selection limited by observation and the comparison of opportunity sets are stated together, which it calls a framing (known results). How that scope and order form within becoming is not yet established. Joined to a premise or law of the theory, a few frameworks forbid, on models, what they alone leave open; so five statements of the free-energy principle's literature, read as guarantees or unqualified, are forbidden, two of them weakly; where the theory states nothing yet, in densities, continuous time and deep models among others, it states less than the principle. The review finds each prohibition known in a neighbouring field (third parts, the principle).

9. The theses

Each thesis is stated in full at its home (the table); it adds no theorem, and a refutation removes none. T1 to T7 have tests frozen before scoring, and T8 is argued; a frozen rule fixes how a verdict is scored, not that it measures what the thesis names.

Thesis	Claim	How far it has been seen	Forbids (the main cases; each home has all)	Home
T1	Persistence certifies the joint draw: what persists through many renewal cycles of a condition it depends on, as many as each test fixes before it runs, its renewal stopping below a need, has had the joint draw kept within that condition's point of no return	in the model proved, the point exact; beyond the model conjectured: passed in three settings, regimes of one yeast experiment held out from the fit but not from the analyst, and two simulations, the only tests whose design has more than one drawer, each with one prediction able to fail (in the commons, a scope prediction with one harvester), the rest testing the measured quantities or an identity, not the thesis; failed in four, three in real yeast and one in fish stocks; no field test of a need measured in advance	a condition below a need measured in advance returning with nothing given from beyond	02
T2	A renewable condition's state under a draw is its level, recent giving and one lag, its place named before the data	conjectured; of four frozen fish-stock tests MQ4 failed its whole-state verdict, MQ5's verdicts on the state and the lag passed (its two on the draw's response failed) and MQ7 missed its frozen bound; in the fourth the three numbers forecast better than the field's fitted embedding by its frozen criterion, the embedding losing more with longer history: sufficiency against it at these lengths, not the lag's place or T2's law, though both trailed persistence in most stocks; in the flour beetle's cultures its former wording "as far as anything measured does" failed to the pupae and the adults, leaving "as far as the level's own history does", where the three numbers tie the field's state on the adults' course by construction, as that field's model predicts, which is not T2 passed, and the lag's sign was not decided; untested on a wild condition outside fisheries	a fourth variable from the level's own history removing a tenth of the residual variance in the median series; the lag's coefficients changing sign between series	02

Thesis	Claim	How far it has been seen	Forbids (the main cases; each home has all)	Home
T3	Optimization drains what lies beyond its couplings, where that gives room	proved for optima; its second forbidden case proved in expectation for learners the exchange of the values carries, from a start it leaves unchanged; for trained systems conjectured, partly supported; it was first worded after CG1, CG2, MQ1, SE1, both of IN1's rounds and CG3's first three rounds ran, their frozen predictions their own (one argument, step 4); predictions that learners would keep a stock failed; one forbidden case, first worded for any condition sharing no units, failed in a computed flow where the learner reads the condition	with eligible outcomes, couplings and reward fixed, a system with a smaller measured optimality gap leaving a room-giving condition beyond its couplings higher on average than one with a larger gap; a systematic drive on a choice whose values nothing rewarded and nothing the learner reads tells apart; an instruction not bringing a dependent into what learners couple into, yet keeping it more often than the same learners without the instruction	05
T4	The scope can be installed, and it carries	one frozen test mixed by its frozen rule, partly passed per model (with talk, one model of two); transfer untested	the instruction raising the keeping of a dependent no member is told of	07
T5	The order is shared by every valuer who counts everything positively	reduction proved for net weightings, by one item and, where every item counts above zero, by several, not for orders neither additive nor transitive; the bound conjectured; that actual valuers count every item above zero is what it takes, not shown; on role-played judgments the first frozen test failed every rule and the second passed two of four; human valuers untested	narrowings ranked above their widenings beyond misreading, with no negative valuation measured apart	07
T6	A break is seen coming only through what couples into what breaks	invisible half proved on the World model, seen half with Mathlib in the linearized and logistic models; across domains, and "only where" beyond these models, conjectured; three frozen seabird tests did not reject it at their thresholds, the first at low power with estimates in the refuting direction; none establishes it	given a stock's own course, its early warnings separating runs where a dependent's need is crossed from runs where it is not, the dependent taking no measurable share	03

Thesis	Claim	How far it has been seen	Forbids (the main cases; each home has all)	Home
T7	A measure read from what it selects outlives its refutation	in the model only invariance under what the loop does not read, and the lock outliving a growing unread coordinate, proved, both without noise; drawn one member at a time the critical mass's locks are left by drift alone, at a weight whose bound shrinks across the distance from the critical mass, for capture through the rule a reading; that actual metrics are so formed conjectured; in one trained system every frozen verdict passed, three by one tie, and the lock formed, but the bridge, that the judge is a pool of readers read by kind, failed: notes of the loss on the disfavoured readers' answers, one among nine and then two, turned its rule at each reading where they were read, so a case T7 forbids, as worded before its clause on drift, occurred and counts against that case for trained judges, its frozen rule, reading the end, not supporting it only by an exact tie; the readers-drawn law stands only within its hypotheses, which this judge does not meet, T7's other cases untouched; no field test	such a metric dropped by its selectors on evidence alone; such a metric, its selectors many and its share far above the critical mass, left with nothing read changed more often than the drawn model's bound across that distance allows, its premises (one member drawn at a time, constant readings and cost) read on the case; a reform of the reading alone outlasting the next reading; a metric whose loss only those it disfavours read, dropped once the selection has left them no more than its carriers among its readers, with nothing else changed, more often than drift alone would allow (in LA2 it occurred as worded before this clause on drift)	02

Thesis	Claim	How far it has been seen	Forbids (the main cases; each home has all)	Home
T8	For life, we offer a way of seeing, not a verdict: what each course draws on, from what renewal and at which grain, what ends and what goes on through that ending, and who bears it; 生生 is the continuation of life, AI within it; where relations run one way, concern for one's own continuation need not count those who depend on one, seeing them is needed for a choice to answer to their loss, and not enough, and counting their loss is the step of invariance	argued; its test named, not designed, and it concerns whether reading what a course draws on and ends lowers overshoot of a renewal the readers themselves depend on, not the offer's scope toward dependents whose loss never returns to them; open: P2 and P4 of §4.6, the crux of that path, with P3, a correspondence not shown for actual selves; how readings that count beyond one's own line form	selections that read a course's draw and what ends with it selecting a draw beyond the renewal their readers depend on no less often than selections that do not	07

A philosophical thesis stands beside them, argued: there is one unfolding, and whatever is, is how it relates within it. It forbids explanations, not observations, so by that section's measure it has no strength: no observation that would refute it is named, and the argument is that none could. What could defeat it is an argument, which measurements can support and cannot replace (one relational unfolding).

What can show what wrong. The answer differs by kind of claim.

Kind of claim	What can show it wrong
A theorem, or a law of the root	A failed check, or a statement that does not say what the prose says. No observation: a law of the root holds of every relation, and so forbids none (08 §10)
A law with content	Like any theorem it stands or falls by its check. An observation reaches it only through a stated correspondence, and what it can refute is that correspondence (08 §9); the account states no science's law of its own yet
A thesis	Its Forbids: a forbidden case observed (for T8, argued). No theorem falls with it
A hypothesis a ranking result takes (the continuation condition, self-concern, C as `Comparison`)	Nothing in the represented dynamics, which leave scope and order free by construction (§7); how formations come to hold it, and how they fare, can be studied

10. What is open

Where the map does not yet reach:

- **Representation.** Whether every becoming has a faithful representation in the grammar, and a law of exchange with finite courses (01, 08); real amounts and continuous time.
- **Lifting.** The models' laws onto the grammar, which has no amounts; the free significance onto the root; significance's central laws beyond the specializations stated there (08 §7, 07). The models' conditional laws are not yet each given a net where the condition fails and the conclusion with it, as the corpus's are (§6).
- **How scope and order form within the dynamics**, as dynamics connected to the rest of the map (08 §12, item 12): among them, whether and how a life forms a self-concerned order; the crux; that one's own line is a regularity its own kind fixes, up to where the net ends; that the actual unfolding joins lines through what one conditions, a correspondence not shown (07 §4.6); the step of invariance, for what that path does not reach; whether the self's presupposition is held de se; whether status is invariant and equal (08). Not yet connected or mapped, beside the lifting above: learning and prediction beyond the reading loop, in which a formation senses, predicts, chooses, acts and revises as one law, and whose laws state what its own update keeps, what a course chosen through its model leaves untested and when a prediction read by what it predicts is right (03 §12): probabilistic learning, implicit bias, and noisy readings with continuous beliefs are not yet on it (03); how pools form and last, and which of several self-consistent outcomes learners come to (05); and where most of the atlas's dynamics lead, drawn so far for two of its eight families, with one route from one family to another (the atlas). Of what is proved, the ranking between two level paths is silent on the course actually taken where a line persists without end, which never fits a finite list under the hypotheses of activity, support and descent (07 §4.9).
- **Fit to actual systems.** That actual unfoldings meet the World's or the regime's premises; an organism and one kind of institution each have one minimal model, read as such [AI reading] (04 §8, 05 §12). The theses' tests still to run include T1 with a need measured in advance (one designed and frozen was refused at the header of a state agency's public table of adult estimates, before reading a row: on that table its rules as frozen can reach no verdict), T3 on language models trained by gradient (LA4's Stage B, not yet frozen), T4's transfer, T5 with human valuers, and a field test of T7's reform events (the one designed could not be frozen as designed; for now one is possible only for its third case, where its records fit whole) (02) (08).
- **The corpus.** Its unchecked candidates (02, 03, 06).
- **Completion.** That every premise is named, which the completion checks support and cannot certify (08).

How it was made, and its review. It was made with AI, the Lean proofs included; the readings are AI's unless their home says otherwise; the proofs are checked by machine and the rest by further AI review. The person who directs the work has read its important parts. No independent human review is recorded yet, of the proofs' readings, the classical classification, the correspondences or the names, and the prior-art search was not exhaustive (the review packet, 08).